

Mainak Debnath

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Education

St. Thomas' College of Engineering and Technology

Kolkata, India

B.Tech in Computer Science and Engineering GPA : 9.08/10

Aug. 2018 – July 2022

Technical Skills

Languages: C#, Typescript, Python, C/C++, SQL, HTML/CSS

Development Frameworks: Angular, .NET, Django, Flask, Langchain

Developer Tools: Git, VS Code, Cursor, Github Copilot, Jupyter notebook, PyCharm, IntelliJ

Libraries: Pandas, NumPy, Matplotlib

Experience

MRI Software

Bengaluru, India

Software Engineer 2

Feb 2025 – Present

- Led the development work for the integration of **SecureSign**, an e-signature solution, into the Affordable Housing product, enabling clients to sign documents digitally and eliminating manual paperwork.
- Accelerated document turnaround by **70%** across Affordable Housing clients by streamlining signature workflows.
- Coordinated cross-functional rollout across environments, ensuring stable deployment and minimal client disruption.
- Collaborated with product, QA, and infrastructure teams to monitor post-deployment success and gather client feedback.

Software Engineer 1

Aug 2022 – Jan 2025

- Built dynamic, responsive UIs for the Affordable Housing platform using **Angular**, improving usability and streamlining user interactions across key application workflows.
- Implemented **lazy loading** and **pagination** for high-volume components, reducing initial load time by **30%** and improving responsiveness.
- Developed scalable backend logic in **.NET**, reducing API latency by **15%** and increasing application throughput.
- Crafted and maintained efficient **SQL queries** for all **CRUD** operations, ensuring data accuracy and integrity across modules.
- Executed rigorous **unit and integration tests** to validate UI/backend behavior, contributing to a stable, bug-resistant codebase.
- Collaborated closely with QA and product teams to triage issues and accelerate sprint velocity by delivering reliable, test-ready features.

Projects

ReviewPal  | Python, LangChain, Gemini AI, ChromaDB, GitHub Actions, GitHub REST APIs

May 2025

- Built an **AI-powered GitHub pull request reviewer** that generates and posts **standards-aware inline feedback** using repository code context and project-specific coding rules.
- Engineered a **CI-triggered RAG pipeline** with **ChromaDB**, combining **semantic retrieval + lexical search**, same-file context prioritization, and **content-hash-based incremental indexing**.
- Improved reviewer quality and runtime through **parallel diff processing**, **second-pass verification**, **confidence-based ranking**, and validation guardrails for added-line accuracy and duplicate suppression.

IntelliBlogger  | Python, DRF, Angular, Redis, Celery, SQLite, Gemini API

Jan 2024 - March 2024

- Built a full-stack AI content platform using **Angular** and **Django REST Framework** to convert YouTube videos into editable long-form blog posts with authenticated user workflows.
- Designed a **job-based generation pipeline** with persisted job state, retry handling, and resumable status tracking to support long-running AI workloads outside the synchronous request-response path.
- Implemented **JWT-based authentication**, user-scoped APIs, endpoint-level **rate limiting**, and **Redis-backed caching** to secure high-cost generation endpoints and reduce repeated generation latency from 25s to under 5s for cached requests.
- Improved backend production readiness through environment-based configuration, health checks, **SSE-based job status updates**, and **Celery**-backed asynchronous processing for long-running generation workflows.

Achievements

- Secured **2nd place** in the annual company-wide hackathon for building **ReviewPal**, an AI-powered automated code review agent integrated with GitHub PR workflows.
- Contributed to various open source projects like **numpy**, **freecodecamp**, **sButtons**.